

Array Methods

- `forEach`
- `map`
- `filter`
- `some`
- `every`
- `reduce`



forEach

```
let arr = [1, 2, 3, 4, 5];
```

```
function print(el) {  
  console.log(el);  
}
```

```
arr.forEach(print);
```

```
// OR
```

```
arr.forEach(function(el) {  
  console.log(el);  
});
```



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JS app.js X

JS app.js >  arr.forEach() callback

```
1 let arr = [1, 2, 3, 4, 5];
2
3 arr.forEach(function (el) {
4   console.log(el);
5 });
6
7 // let print = function (el) {
8 //   console.log(el);
9 // };
10
11 // arr.forEach(print);
12
```

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JS app.js

JS app.js > arr.forEach() callback

```
1  let arr = [  
2    {  
3      name: "aman",  
4      marks: 95,  
5    },  
6    {  
7      name: "shradha",  
8      marks: 94.4,  
9    },  
10   {  
11     name: "rajat",  
12     marks: 92,  
13   },  
14 ];  
15  
16 arr.forEach((student) => {  
17   console.log(student);  
18 })  
19
```



JS

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No Issues

```
▶ {name: 'aman', marks: 95} app.js:17
▶ {name: 'shradha', marks: 94.4} app.js:17
▶ {name: 'rajat', marks: 92} app.js:17
>
```

⏸



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JS app.js X

JS app.js >  arr.forEach() callback

```
1 let arr = [  
2   {  
3     name: "aman",  
4     marks: 95,  
5   },  
6   {  
7     name: "shradha",  
8     marks: 94.4,  
9   },  
10  {  
11    name: "rajat",  
12    marks: 92,  
13  },  
14 ];  
15  
16 arr.forEach((student) => {  
17   console.log(student.marks);  
18 });  
19
```

 marks

(property) marks: number



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95	app.js:17
94.4	app.js:17
92	app.js:17

||



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JS app.js X

JS app.js > [?] double > num.map() callback

```
9      },
10     {
11       name: "rajat",
12       marks: 92,
13     },
14   ];
15
16   // arr.forEach((student) => {
17   //   console.log(student.marks);
18   // });
19
20   let num = [1, 2, 3, 4];
21
22   let double = num.map((el) => {
23     return el * el;
24   });
25
```



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JS app.js X

JS app.js > [?] double > num.map() callback

```
8     marks: 94.4,  
9   },  
10  {  
11    name: "rajat",  
12    marks: 92,  
13  },  
14 ];  
15  
16 // arr.forEach((student) => {  
17 //   console.log(student.marks);  
18 // });  
19  
20 let num = [1, 2, 3, 4];  
21  
22 let double = num.map((el) => {});  
23
```



JS x +

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No Issues

```
> double  
< ▶ (4) [undefined, undefined, undefined, undefined]  
>
```

⏸



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JS app.js



JS app.js > [🔗] gpa > 📦 students.map() callback

```

1  let students = [
2    {
3      name: "aman",
4      marks: 95,
5    },
6    {
7      name: "shradha",
8      marks: 94.4,
9    },
10   {
11     name: "rajat",
12     marks: 92,
13   },
14 ];
15
16 let gpa = students.map((el) => {
17   return el.marks / 10;
18 });
19
20 // arr.forEach((student) => {
21 //   console.log(student.marks);
22 // });
23

```



JS x +

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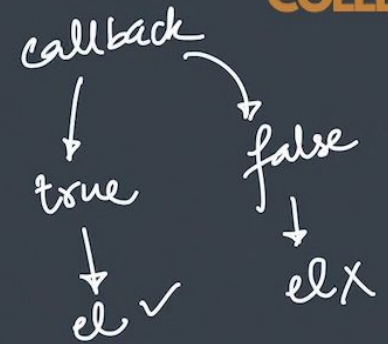
```
> gpa
< ▼ (3) [9.5, 9.440000000000001, 9.2] ⓘ
  0: 9.5
  1: 9.440000000000001
  2: 9.2
  length: 3
  ▶ [[Prototype]]: Array(0)
>
```



Filter

let newArr = arr.**filter**(some function definition or name);

```
let nums = [2, 4, 1, 5, 6, 2, 7, 8, 9];  
  
let even = nums.filter( (num) => (num % 2 == 0) );
```



<> index.html

JS app.js

JS app.js > [x] ans >  nums.filter() callback

```
17   |   return el.marks / 10;
18   | });
19
20   // arr.forEach((student) => {
21   //   console.log(student.marks);
22   // });
23
24   let nums = [1, 2, 3, 4, 7, 8, 2, 9, 10, 12, 11];
25   let ans = nums.filter((el) => {
26   |   return el % 2 == 0;
27   | });
```



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06:50 / 07:57

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JS app.js X

JS app.js > [?] ans > [?] nums.filter() callback

```
17   |   return el.marks / 10;
18   | });
19
20   // arr.forEach((student) => {
21   //   console.log(student.marks);
22   // });
23
24   let nums = [1, 2, 3, 4, 7, 8, 2, 9, 10, 12, 11];
25   let ans = nums.filter((el) => {
26   |   return el % 2 !== 0; //even -> true, odd -> false
27   | });
28
```



JS x +

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No Issues

```
> ans  
< ▶ (5) [1, 3, 7, 9, 11]  
> |
```

⏸



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JS app.js



JS app.js > [x] ans > [x] nums.filter() callback

```
17   return el.marks / 10;
18   });
19
20   // arr.forEach((student) => {
21   //   console.log(student.marks);
22   // });
23
24   let nums = [1, 2, 3, 4, 7, 8, 2, 9, 10, 12, 11];
25   let ans = nums.filter((el) => {
26     return el < 5;
27   });
28
```



JS

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```
> ans  
< ▶ (5) [1, 2, 3, 4, 2]  
>
```



Every

Returns true if **every element of array gives true** for some function. Else returns false.

arr.every(some function definition or name);



```
[1, 2, 3, 4].every( (el) => (el%2 == 0));  
false  
[2, 4].every( (el) => (el%2 == 0));  
true
```



Every 

Returns true if **every element of array gives true** for some function. Else returns false.

`arr.every(some function definition or name);`


true


false 

```
[1, 2, 3, 4].every( (el) => (el%2 == 0));  
false
```

```
[2, 4].every( (el) => (el%2 == 0));  
true
```

JS



JS x +

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```
> [2, 4, 6].every((el) => el%2 == 0);
< true
> [2, 4, 6, 8].every((el) => el%2 == 0);
< true
> [2, 4, 6, 8, 1].every((el) => el%2 == 0);
< false
> [2, 4, 6, 8, 1].every((el) => el%2 != 0);
< false
> [3, 5, 1].every((el) => el%2 != 0);
< true
>
```



Some

OR

Returns true if **some elements of array give true** for some function. Else returns false.

`arr.some(some function definition or name);`

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```
[1, 2, 3, 4].some( (el) => (el%2 == 0));  
true
```

```
[1, 3].some( (el) => (el%2 == 0));  
false
```



Reduce

some
every

single

[1, 2, 3, 4, 5, 6]

Reduces the array to a single value

`arr.reduce( reducer function with 2 variables for (accumulator, element) );`

```
[1, 2, 3, 4].reduce( (res, el) => (res+el) );  
10
```



Reduce

Reduces the array to a single value

some - T/F
every - T/F

single

[1, 2, 3, 4, 5, 6]

`arr.reduce( reducer function with 2 variables for (accumulator, element) );`
el, a, b
↑

```
[1, 2, 3, 4].reduce( (res, el) => (res+el) );  
10
```



Reduce

Reduces the array to a single value

some - T/F
every - T/F

single

[1, 2, 3, 4, 5, 6]

`arr.reduce( reducer function with 2 variables for (accumulator, element) );`

el, a, b

[1, 2, 3]

① acc, el → return
② acc, el - || return
③ acc, el

```
[1, 2, 3, 4].reduce( (res, el) => (res+el) );
10
```

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Reduce

Reduces the array to a single value

some - T/F
every - T/F

single

[1, 2, 3, 4, 5, 6]

`arr.reduce( reducer function with 2 variables for (accumulator, element) );`

el, a, b

① acc, el → return

② acc, el → return ||

③ acc, el → return *Reduced*

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```
[1, 2, 3, 4].reduce( (res, el) => (res+el) );
10
```



$$(0, 1) \Rightarrow \underline{1}$$
$$(1, 2) \Rightarrow \underline{\underline{3}}$$
$$(3, 3) \Rightarrow \underline{\underline{6}}$$
$$(6, 4) \Rightarrow \underline{\underline{10}}$$


$$(0, 1) \Rightarrow 1$$
$$(1, 2) \Rightarrow \underline{\underline{3}}$$
$$(3, 3) \Rightarrow \underline{\underline{6}}$$
$$(6, 4) \Rightarrow \underline{\underline{10}}$$

finalVal



$$(0, 1) \Rightarrow 1$$
$$(1, 2) \Rightarrow \underline{\underline{3}}$$
$$(3, 3) \Rightarrow \underline{\underline{6}}$$
$$(6, 4) \Rightarrow \underline{\underline{10}}$$

finalVal



Reduce

Finding **Maximum** in an array

```
let nums = [2, 3, 4, 5, 3, 4, 7, 8, 1, 2];

let result = nums.reduce( (max, el) => {
  if(el > max) {
    return el;
  } else {
    return max;
  }
});
```



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JS app.js X

JS app.js > ...

```

1  let arr = [1, 4, 2, 5, 6, 7, 2, 9, 2];
2
3  let max = -1;
4
5  for (let i = 0; i < arr.length; i++) {
6      if (max < arr[i]) {
7          max = arr[i];
8      }
9  }
10
11 console.log(max);
12

```

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app.js:11

>

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JS app.js



JS app.js > ...

```
1 let arr = [1, 4, 2, 5, 6, 7, 2, 9, 2];
2
3 // let max = -1;
4
5 // for (let i = 0; i < arr.length; i++) {
6 //   if (max < arr[i]) {
7 //     max = arr[i];
8 //   }
9 // }
10
11 // console.log(max);
12
13 let max = arr.reduce((max, el) => {
14   if (max < el) {
15     return el;
16   } else {
17     return max;
18   }
19 });
20
```



<> index.html

JS app.js

JS app.js > ...

```
1  let arr = [1, 4, 2, 5, 6, 7, 2, 9, 2];
2
3  // let max = -1;
4
5  // for (let i = 0; i < arr.length; i++) {
6  //   if (max < arr[i]) {
7  //     max = arr[i];
8  //   }
9  // }
10
11 // console.log(max);
12
13 let max = arr.reduce((max, el) => {
14   if (max < el) {
15     return el;
16   } else {
17     return max;
18   }
19 });
20
21 console.log(max)
22
```

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Reduce

Finding **Maximum** in an array

```
let nums = [2, 3, 4, 5, 3, 4, 7, 8, 1, 2];  
  
let result = nums.reduce( (max, el) => {  
  if(el > max) {  
    return el;  
  } else {  
    return max;  
  }  
});
```

$[1, 2, 3, 1]$ (max, el)
 $(0, 1) \Rightarrow \underline{1}$
 $(1, 2) \Rightarrow \underline{2}$
 $(2, 3) \Rightarrow \underline{3}$
 $(3, 1) \Rightarrow 3$



Practice Qs

Check if all numbers in our array are multiples of 10 or not.

↳ every

$$\text{num} \% 10 == 0$$

Create a function to find the min number in an array.



Practice Qs

Check if all numbers in our array are multiples of 10 or not.

↳ every

$(num \% 10 == 0)$

↓
true

Create a function to find the min number in an array.



Practice Qs

Check if all numbers in our array are multiples of 10 or not.

↳ every

$(num \% 10 == 0)$

↓
true

Create a function to find the min number in an array.

↓
reduce



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JS app.js X

JS app.js > ...

```
1 let nums = [10, 20, 30, 40, 5];
2
3 let min = nums.reduce((min, el) => {
4   if (min < el) {
5     return min;
6   } else {
7     return el;
8   }
9 });
10
11 console.log(min);
12
```



JS x +

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No Issues

5 app.js:11

>



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JS app.js X

JS app.js > ...

```

1  function getMin(nums) {
2      let min = nums.reduce((min, el) => {
3          if (min < el) {
4              return min;
5          } else {
6              return el;
7          }
8      });
9
10     return min;
11 }
12
13 let nums = [10, 20, 30, 40, 5];
14

```



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JS x +

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No Issues

```
> getMin(nums);  
< 5  
>
```



JS x +

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No Issues

```
> getMin(nums);  
< 5  
> getMin([1, 2, 3,4]);  
< 1  
> getMin([1, 2, 3,4, -1]);  
< -1  
>
```



Default Parameters

Giving a default value to the arguments

```
function func (a, b = 2) {  
    //do something  
}
```

```
function sum(a, b = 3) {  
    return a + b;  
}
```

```
sum(2); //5
```



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JS app.js ×

JS app.js >  sum

```
1  function sum(a, b = 2) {  
2    return a + b;  
3  }  
4
```



JS x +

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```
> sum(1, 4);  
5  
>
```



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> sum(1, 4);

< 5

> sum(1);

< 3

>

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JS app.js

JS app.js > ...

```
1 function sum(a = 2, b) {  
2   return a + b;  
3 }  
4  
5 sum(1, 3); //4  
6 sum(1); //a = 1, b = undefined  
7
```

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JS x +

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```
> sum(1, 3); //4
  sum(1); //a = 1, b = undefined
< NaN

> sum(1, 3); //4
< 4

>
  sum(1); //a = 1, b = undefined
< NaN

>
```

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Spread

iterate

Expands an iterable into multiple values

```
function func (...arr) {  
  //do something  
}
```

```
> let arr = [1, 2, 3, 4, 5];  
< undefined  
> Math.min(...arr);  
< 1  
> console.log(...arr);  
1 2 3 4 5
```

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```
> console.log(..."apnacollege");  
a p n a c o l l e g e
```



JS x +

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```
> Math
< ▶ Math {abs: f, acos: f, acosh: f, asin: f, asinh: f, ...}
> Math.min
< f min() { [native code] }
> Math.min(1, 2, 3);
< 1
> Math.min(1, 2, 3, 1, 2, 3, 0);
< 0
> Math.min(1, 2, 3, 1, 2, 3, 0, 1, 2, 3, 1, 2, 3, 0);
< 0
> let arr = [1, 2, 3, 1, 2, 3, 0, 1, 2, 3, 1, 2, 3, 0];
< undefined
> arr
< ▶ (14) [1, 2, 3, 1, 2, 3, 0, 1, 2, 3, 1, 2, 3, 0]
> Math.min(...arr);
< 0
> arr.push(-1);
< 15
> Math.min(...arr);
< -1
> |
```



JS x +

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```
> console.log(arr);  
  ▶ (15) [1, 2, 3, 1, 2, 3, 0, 1, 2, 3, 1, 2, 3, 0, -1] VM1015:1  
< undefined  
> console.log(...arr);  
  1 2 3 1 2 3 0 1 2 3 1 2 3 0 -1 VM1037:1  
< undefined  
>
```

⏸



JS x +

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```
> console.log(arr);
  (15) [1, 2, 3, 1, 2, 3, 0, 1, 2, 3, 1, 2, 3, 0, -1] VM1015:1
< undefined
> console.log(...arr);
  1 2 3 1 2 3 0 1 2 3 1 2 3 0 -1 VM1037:1
< undefined
> console.log(1, 2, 3, 1);
  1 2 3 1 VM1142:1
< undefined
> console.log("apnacollege");
  apnacollege VM1208:1
< undefined
> console.log(..."apnacollege");
  a p n a c o l l e g e VM1228:1
< undefined
>
```



Spread

with Array Literals

```
> let arr = [1, 2, 3, 4, 5];
```

```
< undefined
```

```
> let newArr = [...arr];
```

```
< undefined
```

```
> newArr
```

```
< ► (5) [1, 2, 3, 4, 5]
```

```
> let chars = [..."hello"];
```

```
< undefined
```

```
> chars
```

```
< ► (5) ['h', 'e', 'l', 'l', 'o']
```



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JS app.js

JS app.js > [🔗] nums

```

1  let arr = [1, 2, 3, 4, 5];
2  let newArr = [...arr];
3
4  let chars = [..."hello"];
5
6  let odd = [1, 3, 5, 7, 9];
7  let even = [1, 4, 6, 8, 10];
8  💡
9  let nums = [...odd, ...even];
10

```



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> nums

< ▶ (10) [1, 3, 5, 7, 9, 2, 4, 6, 8, 10]

>

Spread

with Object Literals

```
let data = {  
  email: "ironman@gmail.com",  
  password: "abcd",  
};  
  
let dataCopy = { ...data, id: 123 };
```



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JS app.js



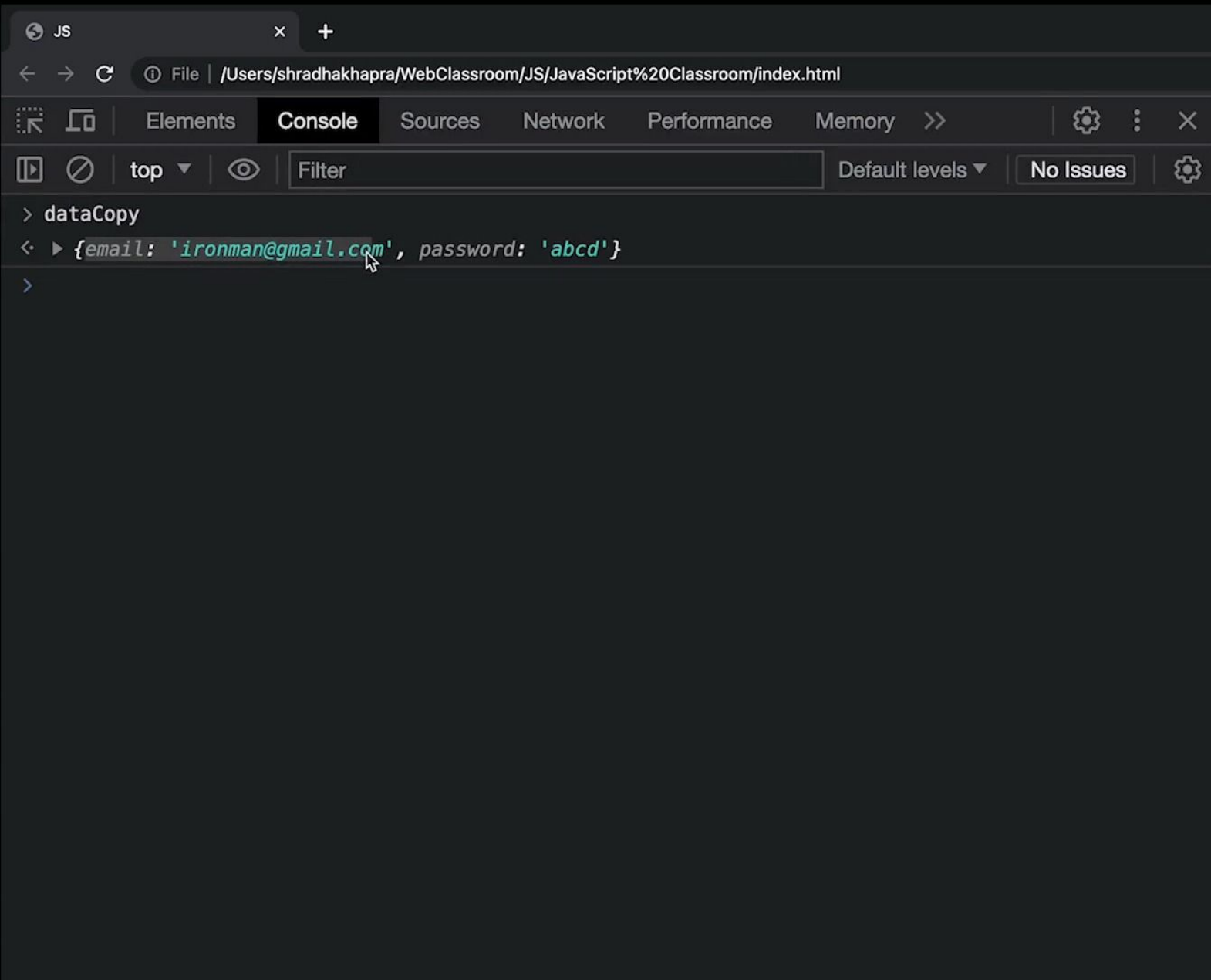
JS app.js > ...

```

1  const data = {
2    email: "ironman@gmail.com",
3    password: "abcd",
4  };
5  💡
6  const dataCopy = {...data};
7
8  // let arr = [1, 2, 3, 4, 5];
9  // let newArr = [...arr];
10
11 // let chars = [..."hello"];
12
13 // let odd = [1, 3, 5, 7, 9];
14 // let even = [2, 4, 6, 8, 10];
15
16 // let nums = [...even, ...odd];
17

```





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JS app.js 1 ●

JS app.js > [🔗] dataCopy

```
1  const data = {  
2    email: "ironman@gmail.com",  
3    password: "abcd",  
4  };  
5  
6  const dataCopy = { ...data, id: 123, country: India };  
7  
8  // let arr = [1, 2, 3, 4, 5];  
9  // let newArr = [...arr];  
10  
11 // let chars = [..."hello"];  
12  
13 // let odd = [1, 3, 5, 7, 9];  
14 // let even = [2, 4, 6, 8, 10];  
15  
16 // let nums = [...even, ...odd];  
17
```



JS x +

← → ↺ ① File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html Press **^⌘F** to exit full screen

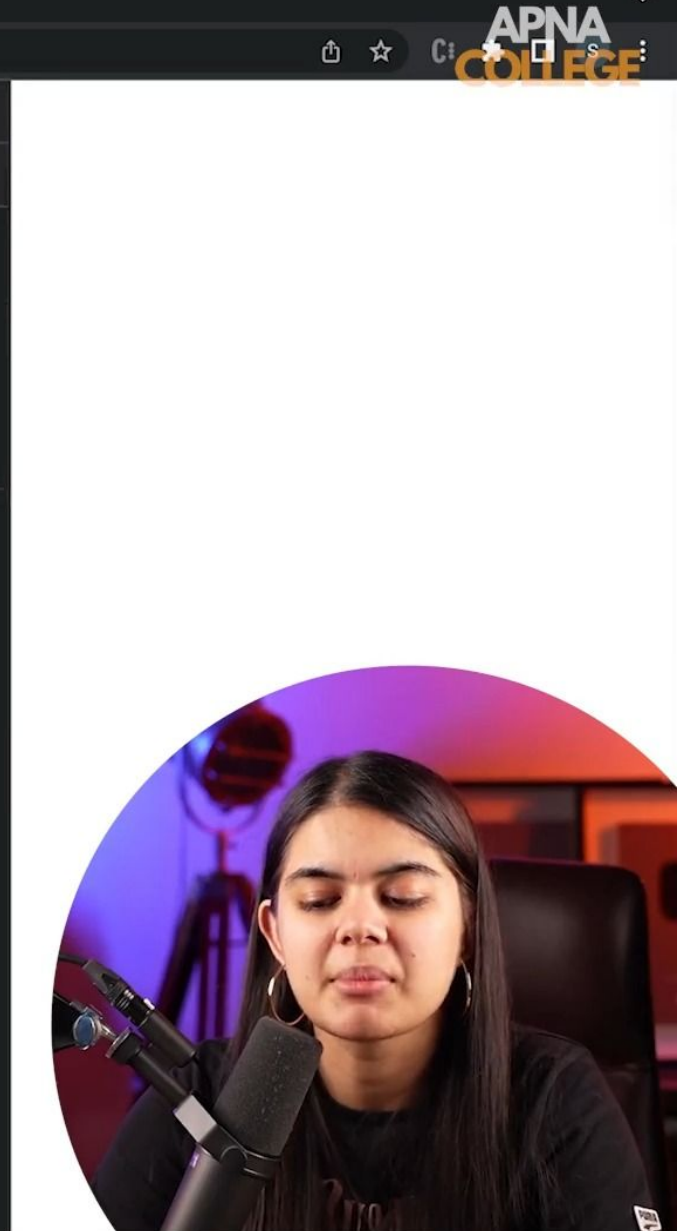
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```
> dataCopy
< {email: 'ironman@gmail.com', password: 'abcd', id: 123, country: 'India'}
  country: "India"
  email: "ironman@gmail.com"
  id: 123
  password: "abcd"
  [[Prototype]]: Object
```

>

⏸



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JS app.js



JS app.js > [obj] obj1

```

1  const data = {
2    email: "ironman@gmail.com",
3    password: "abcd",
4  };
5
6  const dataCopy = { ...data, id: 123, country: "India" };
7
8  let arr = [1, 2, 3, 4, 5]; //val
9  let obj1 = { ...arr }; //obj -> key:val
10
11  // let newArr = [...arr];
12
13  // let chars = [..."hello"];
14
15  // let odd = [1, 3, 5, 7, 9];
16  // let even = [2, 4, 6, 8, 10];
17
18  // let nums = [...even, ...odd];
19

```



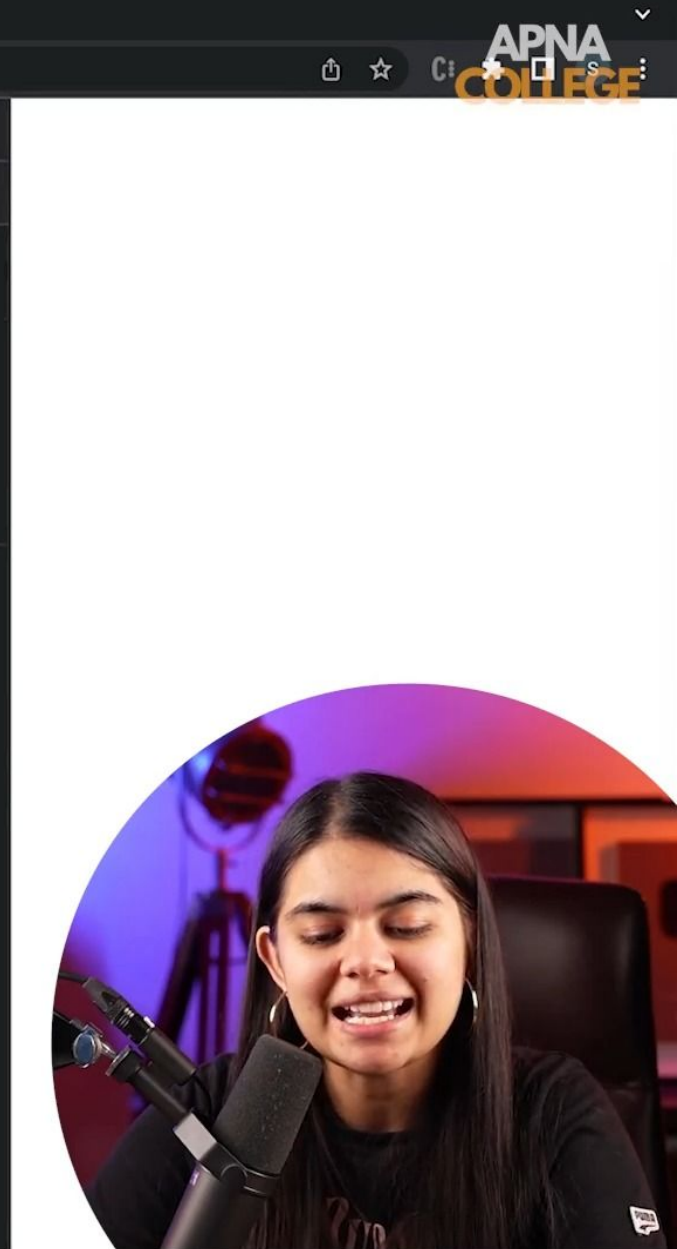
JS x +

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```
> obj1
< {0: 1, 1: 2, 2: 3, 3: 4, 4: 5}
  0: 1
  1: 2
  2: 3
  3: 4
  4: 5
  [[Prototype]]: Object
```



<> index.html

JS app.js X

JS app.js > [obj2] > <unknown>

```

1  const data = {
2    email: "ironman@gmail.com",
3    password: "abcd",
4  };
5
6  const dataCopy = { ...data, id: 123, country: "India" };
7
8  let arr = [1, 2, 3, 4, 5]; //val
9  let obj1 = { ...arr }; //obj -> key:val
10
11 let obj2 = { ..."hello" };
12
13 // let newArr = [...arr];
14
15 // let chars = [..."hello"];
16
17 // let odd = [1, 3, 5, 7, 9];
18 // let even = [2, 4, 6, 8, 10];
19
20 // let nums = [...even, ...odd];
21

```



JS x +

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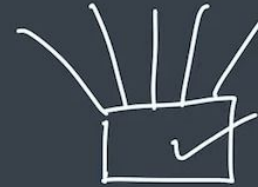
Elements Console Sources Network Performance Memory >> 1

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```
> aobj2
Uncaught ReferenceError: aobj2 is not defined
    at <anonymous>:1:1
VM1605:1

> obj2
{0: 'h', 1: 'e', 2: 'l', 3: 'l', 4: 'o'}
  0: "h"
  1: "e"
  2: "l"
  3: "l"
  4: "o"
  [[Prototype]]: Object
```



Rest $\leftrightarrow$ Spread

Allows a function to take an indefinite number of arguments and bundle them in an array

```
function sum(...args) {  
  return args.reduce((add, el) => add + el);  
}
```



JS x +

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```
> sum(1);
you gave us: 1 app.js:4
< undefined
> sum(1, 2);
you gave us: 1 app.js:4
you gave us: 2 app.js:4
< undefined
> sum(1, 2, 3, 4);
you gave us: 1 app.js:4
you gave us: 2 app.js:4
you gave us: 3 app.js:4
you gave us: 4 app.js:4
< undefined
>
```



<> index.html

JS app.js X

JS app.js > min

```
1 function sum(...args) {  
2   //arguments  
3   for (let i = 0; i < args.length; i++) {  
4     console.log("you gave us: ", args[i]);  
5   }  
6 }  
7  
8 function min(a, b, c, d) {  
9   console.log(arguments);  
10 }  
11
```



JS x +

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top Filter Default levels No Issues

```
> min(1, 2, 3, 4);
  ▶ Arguments(4) [1, 2, 3, 4, callee: f, Symbol(Symbol.iterator): f] app.js:9
< undefined
> |
```

armankhan082020@gmail.com



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JS app.js X

JS app.js > min

```
1  function sum(...args) {  
2    //arguments  
3    for (let i = 0; i < args.length; i++) {  
4      console.log("you gave us: ", args[i]);  
5    }  
6  }  
7  
8  function min() {  
9    console.log(arguments);  
10 }  
11
```

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top Filter Default levels No Issues

```
> min(1, 2, 3, 4);
```

▼ Arguments(4) [1, 2, 3, 4, callee: f, Symbol(Symbol.iterator): f] [app.js:9](#)

- 0: 1
- 1: 2
- 2: 3
- 3: 4
- ▶ callee: f min()
- length: 4
- ▶ Symbol(Symbol.iterator): f values()
- ▶ [[Prototype]]: Object

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```
< undefined
```

```
>
```



<> index.html

JS app.js



JS app.js > min

```
1  function sum(...args) {  
2    //arguments  
3    for (let i = 0; i < args.length; i++) {  
4      console.log("you gave us: ", args[i]);  
5    }  
6  }  
7  
8  function min() {  
9    console.log(arguments);  
10   console.log(arguments.length);  
11   arguments.push(1);  
12 }  
13
```



JS x +

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Elements Console Sources Network Performance Memory >> 1

top Filter Default levels No Issues

```
> min(1, 2, 3, 4);
  Arguments(4) [1, 2, 3, 4, callee: f, Symbol(Symbol.iterator): f] app.js:9
4 app.js:10
Uncaught TypeError: arguments.push is not a function app.js:11
  at min (app.js:11:13)
  at <anonymous>:1:1
```



< index.html

JS app.js X

JS app.js > sum

```
1 function (local var) arguments: IArguments
2 | return arguments.reduce((sum, el) => sum + el);
3 }
4
```



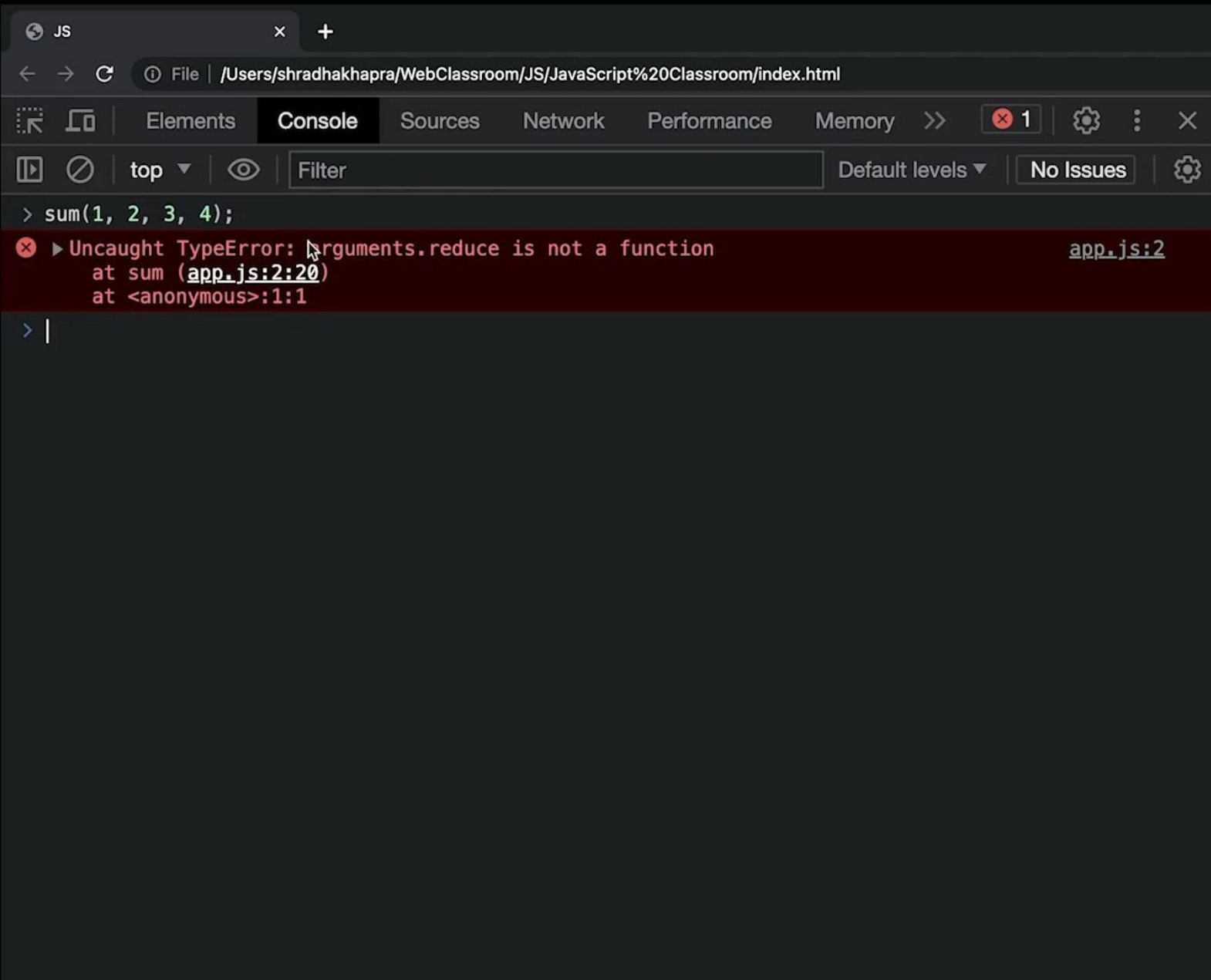
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JS app.js

JS app.js > sum

```
1  function sum() {  
2    return arguments.reduce((sum, el) => sum + el); //a  
3  }  
4
```





< index.html

JS app.js ×

JS app.js > sum

```
1 function sum(...args) {  
2   |   return args.reduce((sum, el) => sum + el);  
3 }  
4
```



<> index.html

JS app.js X

JS app.js > min

```
1  function sum(...args) {
2    return args.reduce((sum, el) => sum + el);
3  }
4  💡
5  function min(...args) {
6    return args.reduce((min, el) => {
7      if (min > el) {
8        return el;
9      } else {
10       return min;
11     }
12   });
13 }
14
```



Auto
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720p
540p
360p
224p

08:56 / 10:31



< index.html

JS app.js X

JS app.js > min

```
1 function sum(...args) {  
2   return args.reduce((sum, el) => sum + el);  
3 }  
4   
5 function min(msg, ...args) {  
6   console.log(msg);  
7   return args.reduce((min, el) => {  
8     if (min > el) {  
9       return el;  
10    } else {  
11      return min;  
12    }  
13  });  
14 }  
15
```

(parameter) args: any[]



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```
> min("hello",12, 445, 123, -20);  
hello app.js:6  
< -20  
> min(10,12, 445, 123, -20)  
10 app.js:6  
< -20  
>
```



Destructuring

Storing values of array into multiple variables

winner

```
let names = ["tony", "bruce", "steve", "peter"];  
let [winner, runnerup] = names;  
console.log(winner, runnerup); // "tony" "bruce"
```



< index.html

JS app.js

JS app.js > ...

```
1 let names = ["tony", "bruce", "peter", "steve"];
2 // let winner = names[0];
3 // let runnerup = names[1];
4 // let secondRunnerup = names[2];
5 💡
6 let [winner, runnerup, secondRunnerup] = names
7
```



< index.html

JS app.js

JS app.js > [e] others

```
1 let names = ["tony", "bruce", "peter", "steve", "abc", "xyz", "pyq"];
2 // let winner = names[0];
3 // let runnerup = names[1];
4 // let secondRunnerup = names[2];
5
6 let [winner, runnerup, ...others] = names;
7
```



JS

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> others

< ▶ (5) ['peter', 'steve', 'abc', 'xyz', 'pyq']

>

Destructuring

Objects

```
const student = {  
  name: "karan",  
  class: 9,  
  age: 14,  
  subjects: ["hindi", "english", "math", "science", "social studies"],  
  username: "karan123",  
  password: 1234,  
};  
  
const { username: user, password: pass } = student;  
  
console.log(user); //"karan123"
```



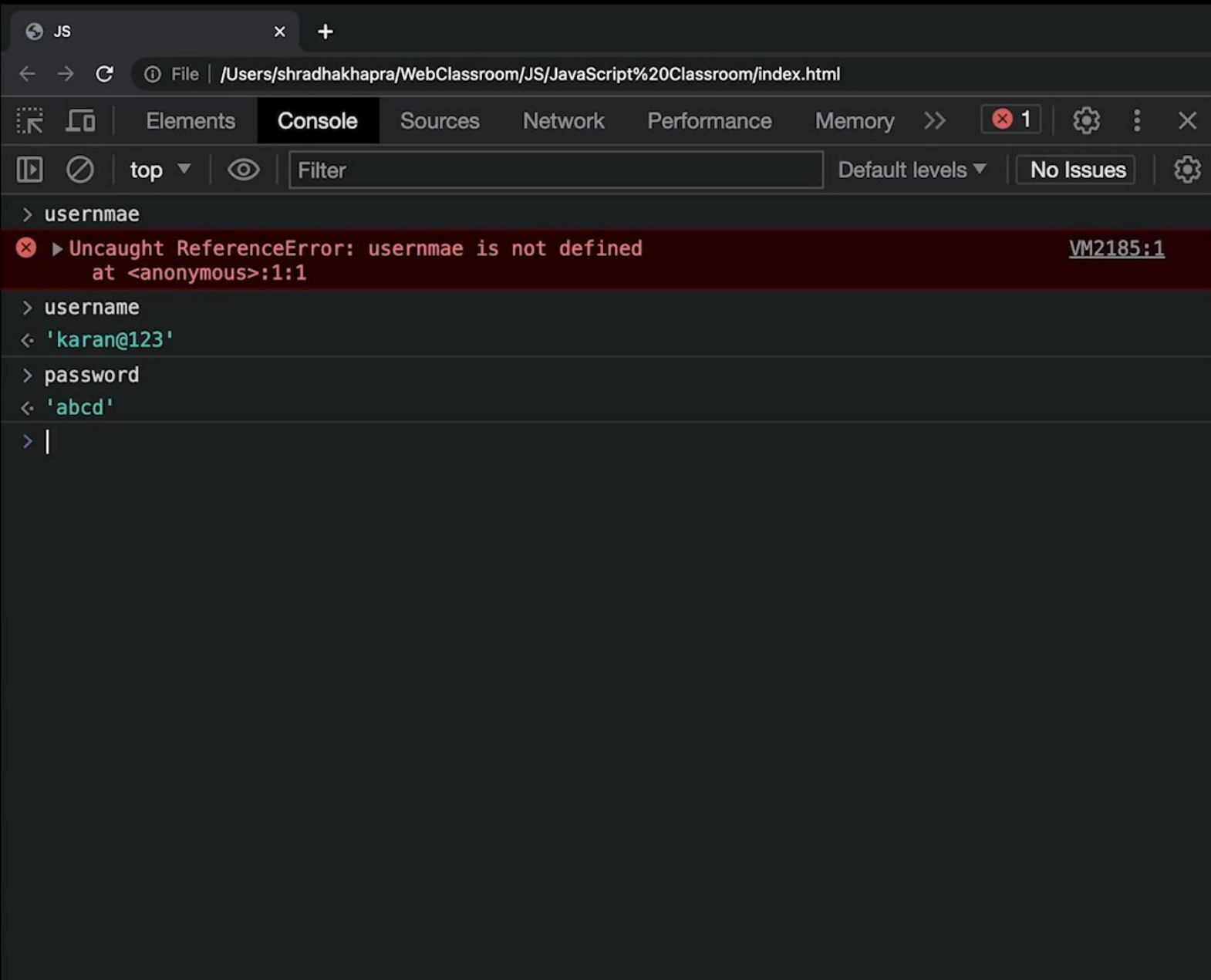
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JS app.js X

JS app.js > [🔗] student

```
1  const student = {
2    name: "karan",
3    age: 14,
4    class: 9,
5    subjects: ["hindi", "english", "math", "science"],
6    username: "karan@123",
7    password: "abcd",
8  };
9
10 let {username, password} = student;
11
```





<> index.html

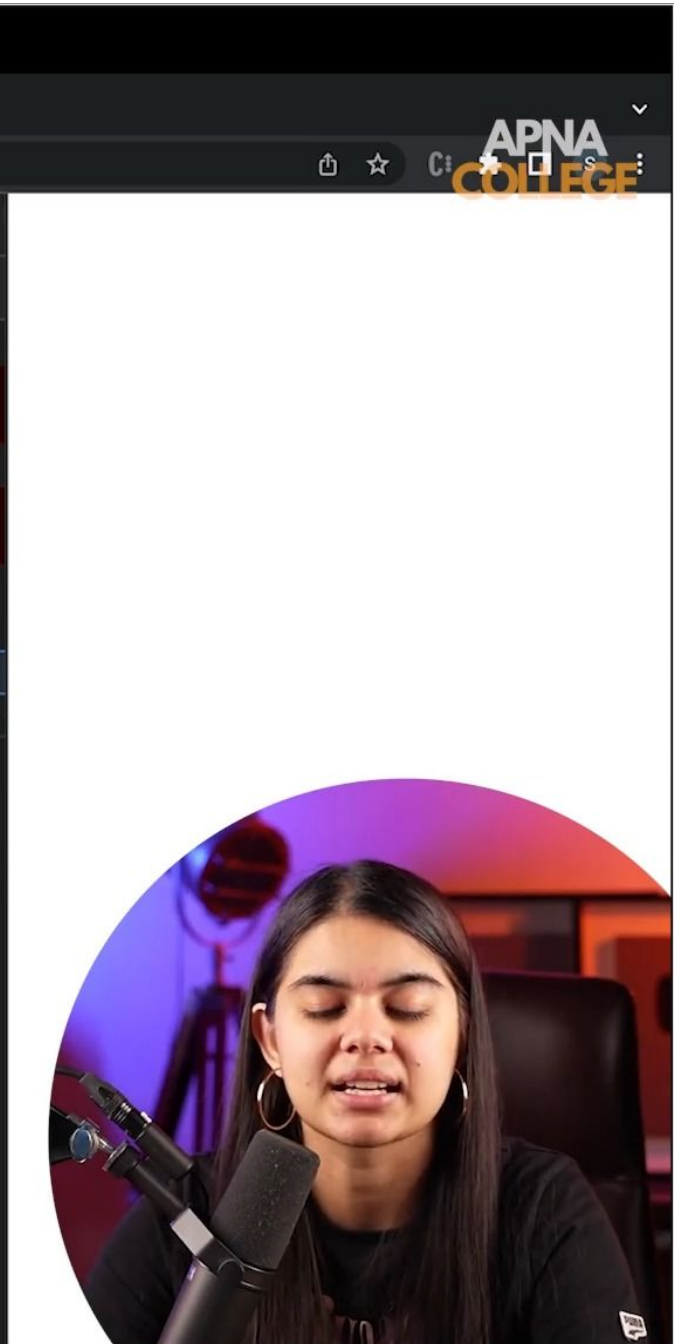
JS app.js



JS app.js > [🔗] secret

```
1  const student = {
2    name: "karan",
3    age: 14,
4    class: 9,
5    subjects: ["hindi", "english", "math", "science"],
6    username: "karan@123",
7    password: "abcd",
8  };
9
10  let { username: user, password: secret } = student;
11
```





<> index.html

JS app.js



JS app.js > [city]

```

1  const student = {
2      name: "karan",
3      age: 14,
4      class: 9,
5      subjects: ["hindi", "english", "math", "science"],
6      username: "karan@123",
7      password: "abcd",
8  };
9
10 let {username: user, password: secret, city = "Mumbai"} = student;
11

```

(property) username: string



JS

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> city

< Mumbai'

>

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<> index.html

JS app.js X

JS app.js > [🔗] city

```
1  const student = {
2    name: "karan",
3    age: 14,
4    class: 9,
5    subjects: ["hindi", "english", "math", "science"],
6    username: "karan@123",
7    password: "abcd",
8    city: "Pune",
9  };
10  💡
11  let { username: user, password: secret, city = "Mumbai" } = student;
12
```



<> index.html

JS app.js



JS app.js > location

```

1  const student = {
2    name: "karan",
3    age: 14,
4    class: 9,
5    subjects: ["hindi", "english", "math", "science"],
6    username: "karan@123",
7    password: "abcd",
8    city: "Pune",
9  };
10  💡
11  let { username: user, password: secret, city: location = "Mumbai" } = student;
12

```



<> index.html

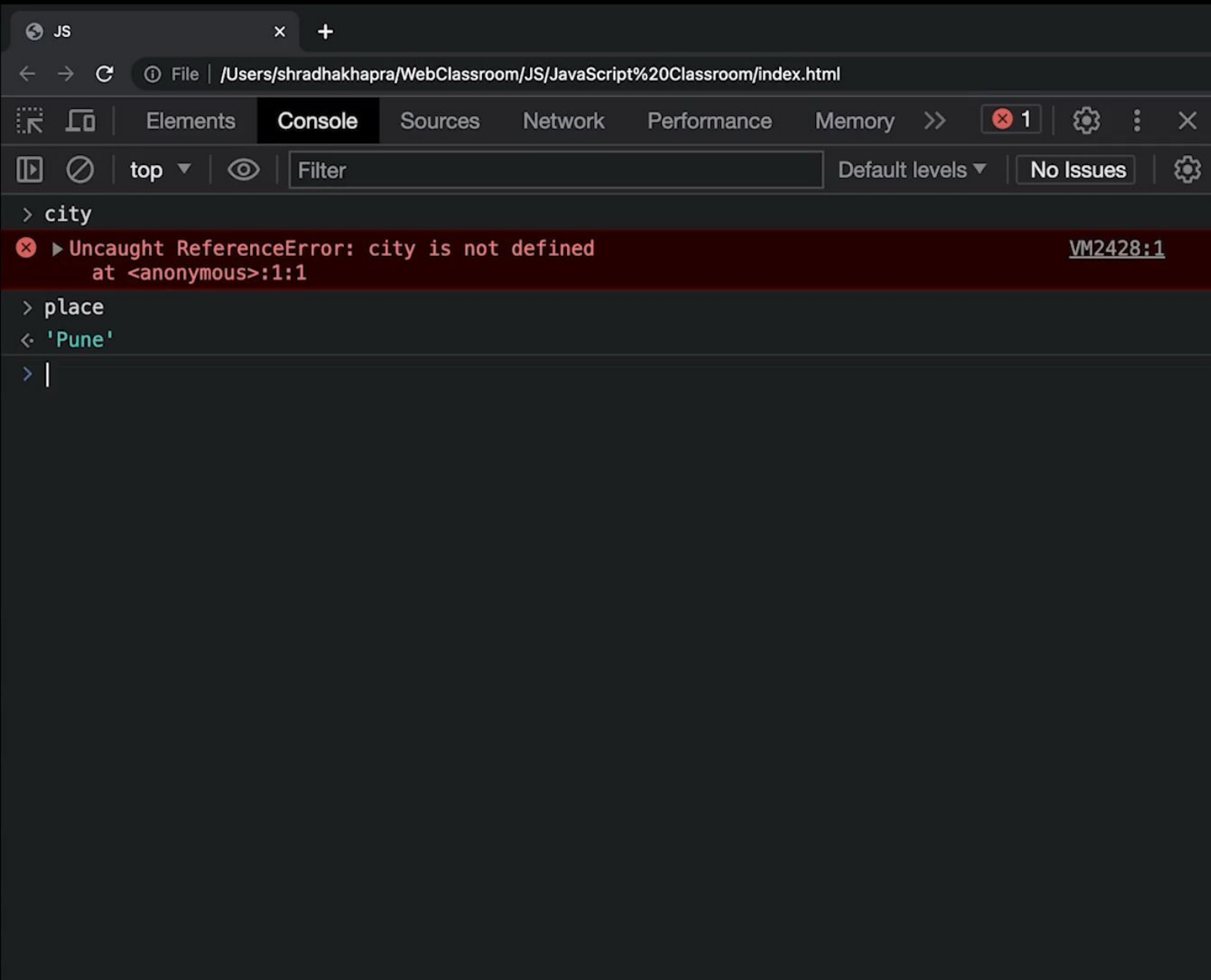
JS app.js



JS app.js > [🔗] place

```
1  const student = {  
2    name: "karan",  
3    age: 14,  
4    class: 9,  
5    subjects: ["hindi", "english", "math", "science"],  
6    username: "karan@123",  
7    password: "abcd",  
8    city: "Pune",  
9  };  
10  💡  
11  let { username: user, password: secret, city: place } = student;  
12
```





<> index.html

JS app.js



JS app.js > [🔗] place

```
1  const student = {  
2    name: "karan",  
3    age: 14,  
4    class: 9,  
5    subjects: ["hindi", "english", "math", "science"],  
6    username: "karan@123",  
7    password: "abcd",  
8  };  
9    
10 let { username: user, password: secret, city: place } = student;  
11
```



JS

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top Filter Default levels No Issues

```
> place
SVGFEDisplacementMapElement is not defined VM2428:1
AnimationPlaybackEvent
> place
< 'Pune'
```

||



JS

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Elements Console Sources Network Performance Memory >>

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No Issues ⚙

> place

< 'Mumbai'

> |



JS (Part 8)

Practice Questions

Qs1. Square and sum the array elements using the arrow function and then find the average of the array.

Qs2. Create a new array using the map function whose each element is equal to the original element plus 5.

Qs3. Create a new array whose elements are in uppercase of words present in the original array.

Qs4. Write a function called `doubleAndReturnArgs` which accepts an array and a variable number of arguments. The function should return a new array with the original array values and all of the additional arguments doubled.

Qs5. Write a function called `mergeObjects` that accepts two objects and returns a new object which contains all the keys and values of the first object and second object.